ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\sqrt{49}$	A rational number can be written as p/q with $q \neq 0$.
2	**(a)** $\sqrt{2}$	The square root of a non-square natural number is irrational.
3	**(a)** 0.75	Divide the numerator by the denominator.
4	**(c)** $0.\overline{3}$	Divide the numerator by the denominator.
5	**(c)** terminating	Recall the basic facts.
6	**(b)** 42	Apply Brahmagupta's rules for signed numbers.
7	**(d)** 14	$14^2 = 196$
8	**(a)** $\frac{7}{9}$	A rational number can be written as p/q with $q \neq 0$.
9	**(a)** $\sqrt{3}$	The square root of a non-square natural number is irrational.
10	**(b)** 0.875	Divide the numerator by the denominator.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Divide 3 by 5 (1); $\frac{3}{5} = 0.6$ (1).	1+1
12	$\sqrt{16} = 4 = \frac{4}{1}$ (1); So it is rational (1).	1+1
13	Use the sign rule for $(-9) \times (-4)$ (1); $= 36$ (1).	1+1
14	Divide 5 by 8 (1); $\frac{5}{8} = 0.625$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let $x = 0.\overline{2}$; $10x = 2.\overline{2}$ (1); $10x - x = 2$, so $9x = 2$ (1); $x = \frac{2}{9}$ (1).	1+1+1
16	Divide 5 by 12 (1); $\frac{5}{12} = 0.41\overline{6}$ (1); Repeats: block 6 (1).	1+1+1
17	Add: $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$ (1); Halve it: $\frac{1}{2} \times \frac{3}{4}$ (1); A rational number between them is $\frac{3}{8}$ (1).	1+1+1
	OR: Let $x = 0.\overline{4}$; $10x = 4.\overline{4}$ (1); $10x - x = 4$, so $9x = 4$ (1); $x = \frac{4}{9}$ (1).	1+1+1
18	Divide 7 by 15 (1); $\frac{7}{15} = 0.4\overline{6}$ (1); Repeats: block 6 (1).	1+1+1
	OR: Add: $\frac{1}{3} + \frac{2}{3} = 1$ (1); Halve it: $\frac{1}{2} \times 1$ (1); A rational number between them is $\frac{1}{2}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sqrt{25}$ is rational (1); $\sqrt{6}$ is irrational (1); $\sqrt{36}$ is rational (1); $\frac{3}{7}$ and 0.6 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Long division gives remainders that repeat (1); $\frac{5}{11} = 0.\overline{45}$ (1); Repeating block 45 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1
20	$\sqrt{2}$ is irrational (1); $\sqrt{49}$ is rational (1); $\sqrt{64}$ is rational (1); $\frac{5}{9}$ and 0.45 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Long division gives remainders that repeat (1); $\frac{7}{33} = 0.\overline{21}$ (1); Repeating block 21 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.
- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt × debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $0.\overline{6}$	Divide the numerator by the denominator.
2	**(c)** non-terminating and non-repeating	Recall the basic facts.
3	**(d)** 3	Apply Brahmagupta's rules for signed numbers.
4	**(c)** 17	$17^2 = 289$
5	**(a)** $0.\overline{3}$	A rational number can be written as p/q with $q \neq 0$.
6	**(a)** $\sqrt{5}$	The square root of a non-square natural number is irrational.
7	**(b)** 0.8	Divide the numerator by the denominator.
8	**(b)** $0.\overline{1}$	Divide the numerator by the denominator.
9	**(a)** rational number	Recall the basic facts.
10	**(d)** -3	Apply Brahmagupta's rules for signed numbers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	7 is not a perfect square (1); So it is irrational (1).	1+1
12	Use the sign rule for $(-18) \div 3$ (1); $= -6$ (1).	1+1
13	Divide 11 by 20 (1); $\frac{11}{20} = 0.55$ (1).	1+1
14	$0.25 = \frac{1}{4}$ (1); So it is rational (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let $x = 0.\overline{5}$; $10x = 5.\overline{5}$ (1); $10x - x = 5$, so $9x = 5$ (1); $x = \frac{5}{9}$ (1).	1+1+1
16	Divide 9 by 11 (1); $\frac{9}{11} = 0.\overline{81}$ (1); Repeats: block 81 (1).	1+1+1
17	Add: $\frac{1}{5} + \frac{3}{5} = \frac{4}{5}$ (1); Halve it: $\frac{1}{2} \times \frac{4}{5}$ (1); A rational number between them is $\frac{2}{5}$ (1).	1+1+1
	OR: Let $x = 0.\overline{7}$; $10x = 7.\overline{7}$ (1); $10x - x = 7$, so $9x = 7$ (1); $x = \frac{7}{9}$ (1).	1+1+1
18	Divide 4 by 9 (1); $\frac{4}{9} = 0.\overline{4}$ (1); Repeats: block 4 (1).	1+1+1
	OR: Add: $\frac{2}{7} + \frac{4}{7} = \frac{6}{7}$ (1); Halve it: $\frac{1}{2} \times \frac{6}{7}$ (1); A rational number between them is $\frac{3}{7}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sqrt{81}$ is rational (1); $\sqrt{11}$ is irrational (1); $\sqrt{12}$ is irrational (1); $\frac{2}{5}$ and 0.125 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Long division gives remainders that repeat (1); $\frac{2}{11} = 0.\overline{18}$ (1); Repeating block 18 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1
20	$\sqrt{13}$ is irrational (1); $\sqrt{100}$ is rational (1); $\sqrt{15}$ is irrational (1); $\frac{7}{8}$ and 0.8 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Long division gives remainders that repeat (1); $\frac{4}{33} = 0.\overline{12}$ (1); Repeating block 12 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.
- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt × debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 19	$19^2 = 361$
2	**(b)** $-\frac{5}{8}$	A rational number can be written as p/q with $q \neq 0$.
3	**(a)** $\sqrt{7}$	The square root of a non-square natural number is irrational.
4	**(c)** 0.45	Divide the numerator by the denominator.
5	**(a)** $0.8\overline{3}$	Divide the numerator by the denominator.
6	**(d)** rational	Recall the basic facts.
7	**(a)** 9	Apply Brahmagupta's rules for signed numbers.
8	**(c)** 23	$23^2 = 529$
9	**(d)** $\sqrt{10}$	The square root of a non-square natural number is irrational.
10	**(a)** 0.52	Divide the numerator by the denominator.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use the sign rule for $-7-(-12)$ (1); $= 5$ (1).	1+1
12	Divide 17 by 25 (1); $\frac{17}{25} = 0.68$ (1).	1+1
13	10 is not a perfect square (1); So it is irrational (1).	1+1
14	Use the sign rule for $(-5)+(-13)$ (1); $= -18$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let $x = 0.\overline{8}$; $10x = 8.\overline{8}$ (1); $10x - x = 8$, so $9x = 8$ (1); $x = \frac{8}{9}$ (1).	1+1+1
16	Divide 13 by 40 (1); $\frac{13}{40} = 0.325$ (1); Terminates (1).	1+1+1
17	Add: $\frac{1}{6}+\frac{1}{3} = \frac{1}{2}$ (1); Halve it: $\frac{1}{2} \times \frac{1}{2}$ (1); A rational number between them is $\frac{1}{4}$ (1).	1+1+1
	OR: Let $x = 0.\overline{1}$; $10x = 1.\overline{1}$ (1); $10x - x = 1$, so $9x = 1$ (1); $x = \frac{1}{9}$ (1).	1+1+1
18	Divide 11 by 30 (1); $\frac{11}{30} = 0.3\overline{6}$ (1); Repeats: block 6 (1).	1+1+1
	OR: Add: $\frac{3}{8}+\frac{5}{8} = 1$ (1); Halve it: $\frac{1}{2} \times 1$ (1); A rational number between them is $\frac{1}{2}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sqrt{144}$ is rational (1); $\sqrt{17}$ is irrational (1); $\sqrt{18}$ is irrational (1); $\frac{9}{4}$ and 0.2 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Long division gives remainders that repeat (1); $\frac{8}{11} = 0.\overline{72}$ (1); Repeating block 72 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1
20	$\sqrt{19}$ is irrational (1); $\sqrt{121}$ is rational (1); $\sqrt{20}$ is irrational (1); $\frac{4}{3}$ and 0.75 are rational (terminating or fraction) (2).	1+1+1+2
	OR: Long division gives remainders that repeat (1); $\frac{13}{33} = 0.\overline{39}$ (1); Repeating block 39 (1); Block length $= 2$ (1); The expansion is non-terminating and repeating, so the number is rational (1).	1+1+1+1+1

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